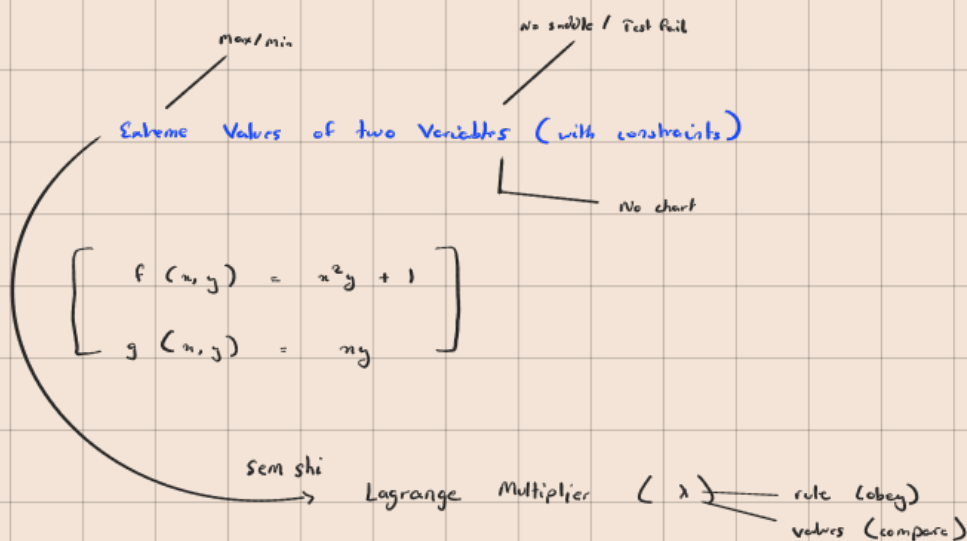




Q) $f(x, y) = xy + 1$

$\therefore$ Taking partial derivatives for

$g(x, y) = x^2 + y^2 - 1$

f_x, f_y, g_x, g_y

$f_x = \lambda g_x$

$f_y = \lambda g_y$

$y = \lambda 2x$

$x = \lambda 2y$

$\lambda = \frac{y}{2x}$

$\lambda = \frac{x}{2y}$

$\lambda = \lambda$

$\frac{y}{2x} = \frac{x}{2y}$

$xy^2 = x^2$

$y = x$

$g(x, y) = x^2 + y^2 - 1$

$\therefore g(x, y) = 0$

$x^2 + y^2 - 1 = 0$

$\therefore y = x$

$x^2 + x^2 - 1 = 0$

$2x^2 - 1 = 0$

$2x^2 = 1$

$x^2 = \frac{1}{2}$

$x = \pm\sqrt{\frac{1}{2}}, y = \pm\sqrt{\frac{1}{2}}$

$(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}), (-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}), (\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$

$\therefore$ plug into $f(x, y)$

- $f(x, y)$ is the main function

- $g(x, y)$ is the sub/constraint

+ve = maximum

-ve = minimum

Q) Find the maximum and minimum

of $f(x, y) = 5x - 3y$ subject

to $g(x, y) = x^2 + y^2 = 136$.

$$f_x = \lambda g_x$$

$$f_y = \lambda g_y$$

$$5 = \lambda 2x$$

$$-3 = \lambda 2y$$

$$\lambda = \frac{5}{2x}$$

$$\lambda = \frac{-3}{2y}$$

$$\lambda = \lambda$$

$$\frac{5}{2x} = \frac{-3}{2y}$$

$$10y = -6x$$

$$y = \frac{-3x}{5}$$

$\therefore$ put in $g(x, y)$

$$\therefore g(x, y) \quad x^2 + y^2 = 136$$

$$x^2 + \left(\frac{9}{25}x^2\right) = 136$$

$$25x^2 + 9x^2 = 136 \times 25$$

$$34x^2 = 3400$$

$$x^2 = 100$$

$$x = \pm 10$$

$$y = \pm 6$$

$$(-10, 6) \quad (-6, 10)$$

$\therefore$ put coords in $f(x, y)$

$$= +68 \quad , \quad -68$$

Max

Min

Q) The temperature T at any point (x, y, z) $T = 400\pi y z^2$.

find the highest temp at the surface of a unit sphere.

$$x^2 + y^2 + z^2 = 1$$

$$f_x = \lambda g_x$$

$$f_y = \lambda g_y$$

$$f_z = \lambda g_z$$

$$200 y z^2 = \lambda 2x$$

$$400 \pi x z^2 = \lambda 2y$$

$$800 \pi y z = \lambda 2z$$

$$\lambda = \frac{200 y z^2}{x}$$

$$\lambda = \frac{200 \pi x z^2}{y}$$

$$\lambda = 400 \pi y$$

$$\lambda_1 = \lambda_2$$

$$\lambda_2 = \lambda_3$$

$$\frac{200 y z^2}{x} = \frac{200 \pi x z^2}{y}$$

$$\frac{200 \pi x z^2}{y} = 400 \pi y$$

$$y^2 = x^2$$

$$200 \pi x z^2 = 400 \pi y^2$$

$$x = y$$

$$z^2 = 2y^2$$

$$x^2 + y^2 + z^2 = 1$$

$$\therefore x = y, \quad z^2 = 2x^2$$

$$x^2 + x^2 + 2x^2 = 1$$

$$4x^2 = 1$$

$$x^2 = \frac{1}{4}$$

$$x = \pm \frac{1}{2}, \quad y = \pm \frac{1}{2}, \quad z = \pm \frac{1}{\sqrt{2}}$$

$$\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right) = +50 \quad \text{Max}$$

$$\left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{\sqrt{2}} \right) = +50 \quad \text{Max}$$

$$\left(\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}} \right) = -50$$

$$\left(\frac{1}{2}, -\frac{1}{2}, -\frac{1}{\sqrt{2}} \right) = -50$$

$$\left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}} \right) = -50$$

$$\left(-\frac{1}{2}, \frac{1}{2}, -\frac{1}{\sqrt{2}} \right) = -50$$

$$\left(-\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}} \right) = +50 \quad \text{Max}$$

$$\left(-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{\sqrt{2}} \right) = +50 \quad \text{Max}$$

Wed, Apr 8

©) find the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid.

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) = 1$$

$$\text{Let } V = (2x)(2y)(2z)$$

$$V_x = \lambda g_x$$

$$V_y = \lambda g_y$$

$$V_z = \lambda g_z$$

$$8xz = \lambda \frac{2x}{a^2}$$

$$8xz = \frac{2y}{b^2}$$

$$8xy = \frac{2z}{c^2}$$

$$\lambda_1 = \frac{4yz a^2}{x}$$

$$\lambda_2 = \frac{4xz b^2}{y}$$

$$\lambda_3 = \frac{4xy c^2}{z}$$

$$\lambda_1 = \lambda_2$$

$$\lambda_2 = \lambda_3$$

$$z = \pm y \frac{c}{b}$$

$$\frac{4yz a^2}{x} = \frac{4xz b^2}{y}$$

$$\frac{4xz b^2}{y} = \frac{4xy c^2}{z}$$

$$z = \pm \frac{x b}{a} \frac{c}{b}$$

$$\cancel{4y^2} / a^2 = \cancel{4x^2} / b^2$$

$$\cancel{4x^2} b^2 = \cancel{4xy^2} c^2$$

$$z = \pm \frac{xc}{a}$$

$$y^2 a^2 = x^2 b^2$$

$$\frac{b^2}{x^2} = \frac{c^2}{y^2}$$

$$ya = \pm xb$$

$$zb = \pm yc$$

$$y = \pm \frac{xb}{a}$$

$$z = \pm \frac{yc}{b}$$

$$z = \pm \frac{xc}{a}$$

$$y = \pm \frac{xb}{a}$$

$$z = \pm \frac{xc}{a}$$

$$\therefore \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) = 1$$

$$\left(\frac{x^2}{a^2} + \left(\frac{xb}{a} \right)^2 \div b^2 + \left(\frac{xc}{a} \right)^2 \div c^2 \right) = 1$$

$$\left(\frac{x^2}{a^2} + \frac{x^2 \cancel{b^2}}{a^2} \times \frac{1}{\cancel{b^2}} + \frac{x^2 \cancel{c^2}}{a^2} \times \frac{1}{\cancel{c^2}} \right) = 1$$

$$= \left(\frac{x^2}{a^2} + \frac{x^2}{a^2} + \frac{x^2}{a^2} \right) = 1$$

$$\frac{3x^2}{a^2} = 1$$

$$3x^2 = a^2$$

$$x^2 = \frac{a^2}{3}$$

$$x = \pm \frac{a}{\sqrt{3}}$$

$$y = \pm \frac{a}{\sqrt{3}} \times \frac{b}{a}$$

$$y = \pm \frac{b}{\sqrt{3}}$$

$$z = \pm \frac{c}{\sqrt{3}}$$

$$\therefore V = (2x)(2y)(2z)$$

$$V = 2 \frac{a}{\sqrt{3}} \times 2 \frac{b}{\sqrt{3}} \times 2 \frac{c}{\sqrt{3}}$$

$$V = \frac{8abc}{3\sqrt{3}}$$

Q) find the max and min values of

$$f(x, y) = y^2 - 4x^2$$

$$\text{subject to } x^2 + 2y^2 = 4$$

$$f_x = \lambda g_x$$

$$f_y = \lambda g_y$$

$$(0, \sqrt{2}) \quad \text{+ve Max}$$

$$-8x = \lambda 2x$$

$$2y = \lambda 4y$$

$$(0, -\sqrt{2}) \quad \text{-ve min}$$

$$-8x - \lambda 2x = 0$$

$$2y - \lambda 4y = 0$$

$$(2, 0) \quad \text{+ve Max}$$

$$-2x(4 + \lambda) = 0$$

$$2y(1 - 2\lambda) = 0$$

$$(-2, 0) \quad \text{+ve Max}$$

$$x = 0, \quad \lambda = -4$$

$$y = 0$$

Tue, Apr 14

9) The temperature T at any point (x, y, z) $T = 400xyz^2$.

find the highest temp at the surface of a unit sphere.

$$x^2 + y^2 + z^2 = 1$$

$$f_x = \lambda_1 g_x$$

$$f_y = \lambda_2 g_y$$

$$f_z = \lambda_3 g_z$$

$$400yz^2 = \lambda_1 \cdot 2x$$

$$400xz^2 = \lambda_2 \cdot 2y$$

$$800xy = \lambda_3 \cdot 2z$$

$$\lambda_1 = \frac{400yz^2}{x}$$

$$\lambda_1 = \frac{200yz^2}{x}$$

$$\lambda_2 = \frac{200xz^2}{y}$$

$$\lambda_3 = \frac{400xy}{z}$$

$$\lambda_1 = \lambda_2$$

$$\lambda_2 = \lambda_3$$

$$\frac{200yz^2}{x} = \frac{200xz^2}{y}$$

$$\frac{200xz^2}{y} = \frac{400xy}{z}$$

$$\frac{y}{x} = \frac{x}{y}$$

$$\frac{z^2}{y} = \frac{2x}{y}$$

$$y^2 = x^2$$

$$z = 2y^2$$

$$y = x$$

$$z^2 = 2y^3$$

$$z^2 = 2x^3$$

$$x^2 + x^2 + z^2 = 1$$

$$z = \sqrt{2}x$$

$$4x^2 = 1$$

$$4x^2 - 1 = 0$$

$$(2x)^2 - 1^2 = 0$$

$$(2x+1)(2x-1) = 0$$

$$2x+1 = 0 \quad 2x-1 = 0$$

$$2x = -1$$

$$x = -\frac{1}{2} \quad x = \frac{1}{2}$$

$$y = -\frac{1}{2} \quad y = \frac{1}{2}$$

$$z = \frac{\sqrt{2}}{2} \quad z = -\frac{\sqrt{2}}{2}$$

$$\left(\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}\right) \quad \left(-\frac{1}{2}, -\frac{1}{2}, -\frac{\sqrt{2}}{2}\right)$$

$$= 450\sqrt{2}, \quad -50\sqrt{2}$$

min

max

